Shiyu Chen 陈时宇

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Research interests: autonomous robotic exploration & active perception · semantic mapping and LiDAR–camera fusion · occlusion-aware next-best-view planning · embodied intelligence & VLM-in-the-loop industrial inspection

EDUCATION

Carnegie Mellon University — M.S. in Mechanical Engineering (Research) Aug 2025 – May 2027 (expected)
Computational Engineering and Robotics Lab (CERLAB), advised by Prof. Kenji Shimada Pittsburgh, PA

- Focus: robot autonomous semantic exploration and 6-DoF active inspection for industrial environments.

Univ. of Cincinnati & Chongqing University — Joint Program, B.S. / B.Eng. in Mechanical Engineering 2020 – 2025
Robotics Minor (UC) · UC GPA 3.71/4.0 · CQU GPA 3.58/4.0 · Dean's List ×4 Cincinnati, OH / Chongqing, CN

RESEARCH EXPERIENCE

Graduate Researcher, CERLAB Roboteam (CMU × YKK factory-automation inspection) Aug 2025 – Present

Real-time LiDAR–camera semantic registration on a legged platform

- Built a real-time pipeline fusing Ouster LiDAR geometry with Insta360 X4 panoramic imagery on a Boston Dynamics Spot, producing dense semantically-colored point clouds for downstream mapping and planning.
- Achieved **10 FPS at full LiDAR rate** with **<100 ms latency** and **<10 px reprojection error**, keeping spatial consistency under high-frequency gait vibration; designed a 3D-printed rigid sensor mount for stable extrinsic calibration.

6-DoF occlusion-aware active viewpoint planning (current focus)

- Proposed a Macro–Micro framework where global 2D-frontier exploration is interrupted by **VLM-detected semantic blind spots**, triggering local 6-DoF body-pitch adjustments to recover FOV beyond the LiDAR's ±22.5° vertical limit in narrow corridors and around tall equipment.
- Developing evaluation in NVIDIA Isaac Sim 5.0 (ROS2, Dockerized) with metrics for geometric coverage (ICP volume overlap), semantic completeness, and exploration efficiency; kinematic-first Spot pose control for rapid iteration.

Undergraduate Researcher, Cable-Driven Soft Robotic Arm for Minimally Invasive Surgery Sep 2024 – May 2025
Univ. of Cincinnati, advised by Prof. Janet (Jiangxia) Dong

- Designed and prototyped a wire-actuated continuum arm with **0.27 mm/step** resolution and **>230° bending range**; modeled nonlinear actuation and led hardware–control integration (motor selection, microstepping, Ecoflex validation) with clinical end-users; published in *ASME J. Medical Diagnostics* (2026).

Research Assistant, Deep-Learning Defect Detection in Composite Materials Jan 2022 – Jul 2022
Chongqing University, advised by Prof. Bo Yang

- Developed CNN models (LeNet/VGG16) with multi-scale feature extraction and DSP-based analysis for defect detection/localization in carbon-fiber and welded parts, reaching **>97% accuracy**; applied CenterNet/EfficientDet on image datasets.

PUBLICATIONS & PATENTS

Y. Wang, C. Liu, K. Jiang, B. Wu, **S. Chen**, J. Dong, A. Ashfaq, "Innovative Flexible Robotic Arm for Enhanced Precision in Transaortic Surgical Myectomy," *ASME Journal of Medical Diagnostics*, 9(2): 021001, 2026.

S. Chen, "DC Motor Control in Noisy Environments: A Comparative Study of Wavelet MRPID vs. CNN-Attention Integrated PID," *Int. Conf. on Computer Science and Mechatronics (ICCSM)*, Shanghai, 2024.

L. Liu, **S. Chen**, Z. Tang, "Intelligent Diagnosis of Bearing Faults Based on SDP Image Fusion," *Int. Conf. on Image Processing, Computer Vision and Machine Learning (ICICML)*, Chengdu, 2023.

Patents (China, as co-inventor): 4 granted invention patents (2025–2026) — CN202410147975.0, CN202311182193.2, CN202311183266.X, CN202211167725.0: deep-learning machine vision for steelmaking process monitoring (steel-stream landing-point detection, slag-splashing recognition, slag-discharge state detection); 3 pending applications (CN202410682909 · CN202211704844.4 · CN202211422884) in belt-conveyor inspection.

INDUSTRY EXPERIENCE

Computer Vision R&D Assistant, CISDI Information Technology May 2022 – Dec 2023
Chongqing, China

- Trained/optimized PyTorch detection models for an industrial belt-conveyor inspection system and deployed them to embedded edge devices (NVIDIA Xavier NX / AGX Orin) via TensorRT, cutting inference latency by **40%**; ported Python algorithms to C++ for embedded targets.
- Adapted models to Huawei Ascend (2nd prize, Ascend AI competition); built a PyQt annotation tool and led labeling of 100k+ images. Received return offer.

TECHNICAL SKILLS

Programming	C++ (proficient), Python (proficient), MATLAB, C
Robotics / Sim	ROS1 & ROS2, NVIDIA Isaac Sim, Spot SDK, LiDAR–camera calibration, frontier exploration, SLAM, TSP/path planning (LKH)
Perception / ML	PyTorch, TensorRT, YOLOv7, SAM2, VLM APIs, CNN/deep learning, point-cloud processing (ICP)
Tools	Linux, Docker, Git, SolidWorks, COMSOL

HONORS & TEACHING

- **National Bronze Prize**, 8th China Int'l "Internet+" Innovation & Entrepreneurship Competition (Push.AI — knowledge-graph decision platform for smart manufacturing), 2023.
- Teaching Assistant — Engineering Design Thinking II (UC, 2025) and C++ Programming (CQU, 2024).